A Prospective Randomized Study of Anterior Cruciate Ligament Reconstruction

A Comparison of Patellar Tendon and Quadruple-Strand Semitendinosus/Gracilis Tendons Fixed With Bioabsorbable Interference Screws

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Background: Debate exists regarding the optimal graft for anterior cruciate ligament reconstruction. Few studies have compared the differences in outcome after reconstruction using similar fixation methods.

Hypothesis: Similar outcomes will be seen after anterior cruciate ligament reconstruction with bone–patellar tendon–bone or quadruple–strand semitendinosus/gracilis tendons fixed with bioabsorbable interference screws.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: Ninety-nine patients were prospectively randomized to bone–patellar tendon–bone (46 patients) or quadruple-strand semitendinosus/gracilis (53 patients) reconstruction groups. The bone–patellar tendon–bone group had slightly lower preinjury Tegner scores (6.7 vs 7.1, $P = .03$); otherwise, the groups were similar. All surgeries were performed by a single surgeon using an endoscopic technique with bioabsorbable interference screw fixation. Patients were evaluated at 3, 6, 12, and 24 months.

Results: Forty-six bone–patellar tendon–bone and 50 quadruple-strand semitendinosus/gracilis patients were available at 24 months (97%). No differences in International Knee Documentation Committee grade, Lysholm score, Tegner activity level, range of motion, single-legged hop test, KT-1000 arthrometer manual maximum difference, Short Form-36, or patient knee rating were found. The bone–patellar tendon–bone group had better flexion strength in the operated leg than in the nonoperated leg (102% vs 90%, $P = .0001$), fewer patients complaining of difficulty jumping (3% vs 17%, $P = .03$), and a greater number of patients returning to preinjury Tegner level (51% vs 26%, $P = .01$). The quadruple-strand semitendinosus/gracilis group had better extension strength in the operated leg than in the nonoperated leg (92% vs 85%, $P = .04$), fewer patients with sensory deficits (14% vs 83%, $P = .0001$), and fewer patients with difficulty kneeling (6% vs 20%, $P = .04$). Both groups showed significant improvement in KT-1000 arthrometer manual maximum difference, Lysholm score, Tegner activity level, International Knee Documentation Committee grade, and patient knee rating score.

Conclusions: Good outcomes were seen in both the bone–patellar tendon–bone and quadruple-strand semitendinosus/gracilis groups. Subtle differences were noted between the groups, which may help guide optimal graft choice.

Keywords: anterior cruciate ligament (ACL); patellar tendon: hamstring tendon; ligament reconstruction; clinical trial; bioabsorbable screw

Anterior cruciate ligament (ACL) reconstruction is one of the more commonly performed orthopaedic surgery procedures in the United States. It is most frequently performed with either bone–patellar tendon–bone (BPTB) or quadrupled semitendinosus and gracilis tendon (QSTG) autograft, but controversy exists as to which graft is best. An ideal graft is one that is strong, heals rapidly, has minimal morbidity, and most importantly provides a functionally stable knee. In vitro studies have shown each of these

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autograft choices to be of adequate strength and stiffness compared with the native ACL.[9,16,45,46] Although animal studies suggest that the BPTB graft heals more rapidly with its bone-to-bone interface,[31] the soft tissue–to-bone interface of the QSTG graft is healed by 9 to 12 weeks.[43] One perceived drawback to using the BPTB graft is that it has often been associated with increased anterior knee pain and crepitus.[1] With an accelerated rehabilitation program, some of these issues may be eliminated.[38] Each graft has its proponents, and both graft sources can provide good results.[4,19,32,37]

A number of studies have compared these 2 graft choices. The initial study by Marder et al[26] showed no significant difference in outcomes other than weakness in hamstring torque in the hamstring tendon autograft group. Aglietti et al[1] and Otero and Hutcheson[30] showed better results with patellar tendon grafts with respect to stability. Aglietti et al[1] also showed a greater ability to return to sports. Three of the 4 meta-analyses in the literature have reported better stability with patellar tendon grafts.[14,15,33,47] A systematic review of 9 prospective randomized studies comparing the 2 graft choices was undertaken by Spindler et al.[39] They found that 3 of 7 studies showed better stability with the patellar tendon grafts, and 4 of 4 studies reported greater kneeling pain with the patellar tendon graft.

One of the difficulties in comparing these studies is that frequently different fixation methods were used for different graft types. In most cases, the patellar tendon graft was fixed with interference screws for a more anatomic aperture fixation, while the hamstring grafts were typically fixed with some type of suspensory fixation, such as staples, sutures and a post, or a screw and washer. With the variations in fixation, it is difficult to determine whether the differences in stability are a result of the method of fixation or due to the actual graft. With the advent of bioabsorbable interference screws, it has become more common to use this type of interference aperture fixation for both patellar tendon and hamstring grafts.

The purpose of this study was to compare, in a prospective randomized fashion, the outcomes after ACL reconstruction using BPTB and QSTG grafts fixed with bioabsorbable interference screws. Our hypothesis was that similar stability and functional outcomes would be obtained using either BPTB or QSTG as the graft source when both grafts were fixed using bioabsorbable interference screws.

METHODS

Between January 2000 and October 2003, all patients scheduled for ACL reconstruction who had met the inclusion criteria of the senior author's (GBM) clinic were invited to participate in a randomized trial comparing ACL reconstruction with either a BPTB or QSTG graft. Patients were included if they had a chronic unilateral ACL-deficient knee confirmed by physical examination and magnetic resonance imaging (MRI) scan and were experiencing instability, or if they had clinical and MRI evidence of an ACL tear and planned to return to high-risk sports. Exclusion criteria included open physes, previous ligament surgery to either knee, previous ligament injury to the contralateral knee, Outerbridge grade IV chondral injury >1 cm^2 found at arthroscopy, or multiple ligament injuries other than a grade I or II medial collateral ligament tear.

Patients were counseled before making their decision that the literature suggested that good results could be obtained with either graft. They were informed that some studies had favored the BPTB for stability and the QSTG for decreased kneeling pain, but the overall outcomes for both grafts were similar. The incisions for each graft harvest site were described to the patients. Patients were then free to decide if they were willing to participate in the study and undergo randomization for the graft type. The study was approved by the Institutional Review Board, and all patients gave informed consent.

During this time period, a total of 242 patients were scheduled for surgery. One hundred fifteen patients did not want to be randomized; 71 of those patients chose BPTB, 38 chose QSTG, and 6 chose allograft. Twenty-seven patients were excluded because they did not meet the inclusion criteria. One hundred patients met the inclusion criteria and agreed to participate in the study. Patients were randomized to either BPTB or QSTG graft at the time of surgery using a computerized random number generator. One patient was noted to have a grade IV chondral lesion >1 cm^2 at the time of surgery and was therefore excluded, leaving 99 patients: 46 patients randomized to the BPTB group and 53 patients to the QSTG group.

Both groups were similar with respect to age, sex, involved leg, time to surgery, knee rating, and intraoperative findings (Table 1). Preinjury Tegner scores differed, with the BPTB group being slightly less active than the QSTG group (6.8 vs 7.2, P = .03); however, the preinjury International Knee Documentation Committee (IKDC) activity scores and preoperative Lysholm scores did not show any difference between the 2 groups. No difference was found in anterior knee stability as measured with the KT-1000 arthrometer (MEDmetric, San Diego, Calif) at 134 N (30 lb) and manual maximum. Interestingly, we did find that a majority of the patients in each group complained of either mild, moderate, or severe kneeling pain before surgery (BPTB of 82% and QSTG of 81%), but there was no difference noted between the groups.

The group that agreed to participate in the study had preinjury Tegner level similar to that of the nonstudy group (study of 7.0 vs nonstudy of 6.9, P = .2). When the groups were broken down by graft type, significant differences were noted (study BPTB of 6.8 vs nonstudy BPTB of 7.5, P = .002; and study QSTG of 7.2 vs nonstudy QSTG of 5.9, P = .01).

Surgical Technique

All surgeries were performed by the same surgeon using an arthroscopic single-incision technique. Grafts were harvested based on the randomization performed in the operating room. All grafts were fixed with poly L-lactide bioabsorbable interference screws (Arthrex, Naples, Fla) for both the femoral and tibial fixation.

TABLE 1
Preoperative Comparisons of Patellar Tendon (BPTB) and Hamstring Tendon (QSTG) Groups[a]

Characteristic	BPTB (n = 46)	QSTG (n = 53)	*P* Value
Age (y)			.88
Mean	27.2	27.7	
Range	15-42	14-48	
Sex			.06
Male	31	45	
Female	15	8	
Involved extremity			.23
Right	25	22	
Left	21	31	
Time to surgery			.34
0-3 mo	11	9	
3.1-6 mo	14	12	
>6 mo	21	32	
Tegner preinjury score (0-10)	6.8	7.2	.03
Tegner preoperative score (0-10)	2.7	2.9	.72
Lysholm preoperative score (0-100)	63.6	66.5	.42
IKDC preinjury activity score			.41
Strenuous	35	45	
Moderate	8	6	
Light	3	2	
Sedentary	0	0	
Meniscus			
Medial meniscal tear	28	36	.46
Medial meniscal repair	18	22	.81
Lateral meniscal tear	23	32	.30
Lateral meniscal repair	7	14	.17
KT-1000 arthrometer difference (mm)			
30 lb	3.4	3.1	.74
Manual maximum	7.2	6.5	.36
Knee rating (0-100)	50.0	54.3	.25
Kneeling pain			.23
None	8	10	
Mild	20	20	
Moderate	16	16	
Severe	1	7	

[a]BPTB, bone–patellar tendon–bone; QSTG, quadrupled semitendinosus and gracilis tendon; IKDC, International Knee Documentation Committee.

BPTB Grafts. A 6- to 7-cm incision was made over the medial aspect of the patellar tendon. The paratenon was incised, and a 10-mm-wide section of the middle third of the patellar tendon was incised with a 20-mm bone block from the patella and a 25-mm block from the tibial tubercle. The bone blocks were prepared to fit through a 9-mm sizing tunnel for the patellar bone block and a 10-mm tunnel for the tibial bone block. After the graft preparation was complete, the graft was placed under tension on the back table. At the end of the procedure, the patellar defect was bone-grafted with the excess bone trimmed from the bone blocks, and the paratenon was repaired. As the graft was being prepared by an assistant, the knee was arthroscoped and meniscal injuries were treated as needed. A notchplasty was performed to adequately visualize the over-the-top position. A posterior cruciate ligament-referencing tibial guide (Arthrex) set at 55° was used to place a tibial guide pin into the posterior half of the tibial ACL attachment site. The knee was brought into full extension to evaluate the pin position in relation to the femoral notch. If the pin position was appropriate, it was reamed to a 10-mm diameter and dilated to 10.5 mm to facilitate passage of the tibial bone block. A 6-mm femoral over-the-top guide (Arthrex) was placed transtibially to the over-the-top position at the 1-o'clock (left knee) or 11-o'clock (right knee) position. A guide pin was placed, and a 9-mm reamer was used to outline the femoral tunnel site. The guide pin was positioned so that approximately 1 mm of back wall would be remaining. The femoral tunnel was drilled to a depth of 30 mm. The graft was passed, and the femoral side was fixed with a 7 × 23-mm bioabsorbable screw placed over a guide wire. The knee was flexed to 100°, and the guide wire was passed through a low medial accessory portal to ensure that the screw would be collinear with the graft. The knee was cycled approximately 20 times, and then the tibial side was fixed with the knee in 20° of flexion with manual tension applied to the tibial sutures and a posterior drawer applied. The tibial side was fixed with a 9 × 28-mm bioabsorbable interference screw.

QSTG Grafts. A 3-cm incision was made 5 cm below the medial joint line and 2 cm medial to the tibial tubercle. The sartorial fascia was incised and the semitendinosus and gracilis tendons were removed using a closed-end tendon stripper (Arthrex). The sartorial fascia was then repaired. Excess muscle was removed from the tendons, and they were cut to a length of 22 cm. The grafts were then doubled over a No. 5 passing suture, and the free ends of the semitendinosus and gracilis tendons were each whip-stitched together with a No. 2 suture for a length of 30 mm. All 4 tendons were whip-stitched together at the proximal end for a distance of 30 mm. The graft was sized to the nearest 0.5 mm.

Bone tunnels were drilled as previously described, except that the diameter of the tunnel was reamed at least 1 mm smaller than the graft diameter and then dilated to the same diameter as the graft. A screw that was 0.5 to 1 mm larger than the graft was used for femoral fixation. The knee was cycled approximately 20 times, and then the tibial side was fixed with the knee in 20° of flexion with manual tension applied to the tibial sutures and a posterior drawer applied. The tibial fixation was performed with 2 bioabsorbable screws, the first oversized by 0.5 to 1 mm and 20 mm in length, which was placed anterior to the graft at the level of the subchondral bone of the tibial plateau. The second screw was oversized by 1.5 to 2 mm and was 17 mm in length. This screw was placed posterior to the graft and inserted so that it maintained contact with the tibial cortex at the external tibial aperture (Figure 1).

Rehabilitation. All surgeries were performed on an outpatient basis. The postoperative protocol for both groups was identical and was directed by the same therapist. A hinged knee brace and crutches were used for 4 weeks. Patients were allowed full weightbearing, with the brace

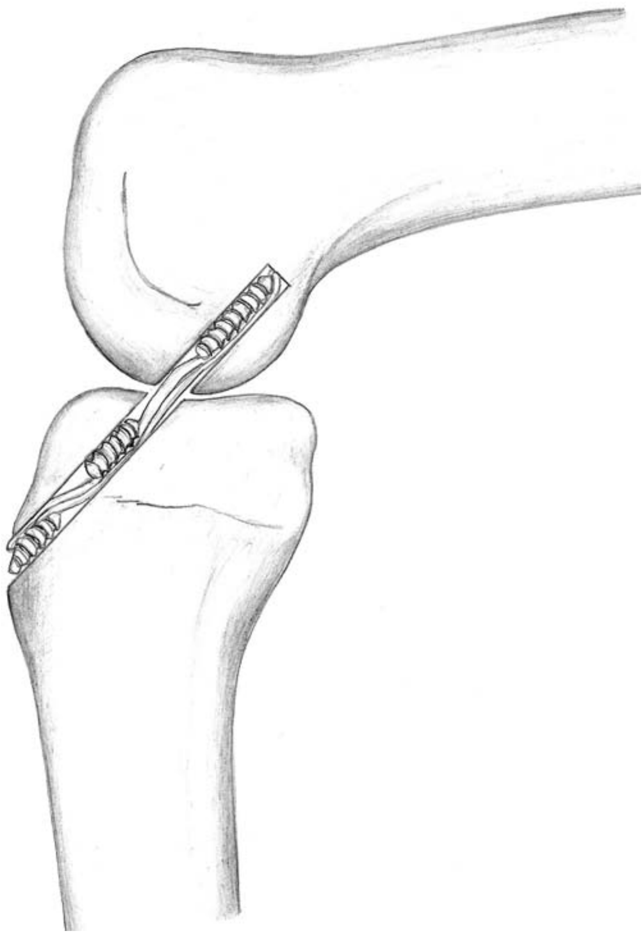

Figure 1. Quadrupled semitendinosus graft fixation with 2 bioabsorbable screws used to secure the tibial side. The proximal screw is 20 mm in length and placed anterior to the graft at the subchondral bone of the proximal tibia. The distal screw is 17 mm in length and placed posterior to the graft at the tibial cortex.

locked in extension, and full active and passive range of motion. Patients who underwent concomitant meniscal repair were allowed partial weightbearing in extension, which was gradually increased to full weightbearing by 4 weeks, and range of motion that was limited to 90° for 4 weeks. After 4 weeks, brace and crutch use was discontinued.

Postoperative exercises, which were begun the same day as surgery, included knee flexion, knee extension with the heel elevated on a rolled towel, and isometric quadriceps muscle contractions. Special emphasis was placed on achieving extension equal to the nonoperated side. Formal physical therapy was begun 1 to 2 weeks after surgery. Patients were scheduled for therapy twice a week for a minimum of 3 months. Patients were not charged for therapy visits in order to encourage maximal participation. After 3 months, patients were transitioned to a home program with periodic follow-up to monitor progress. Stationary bike exercises were encouraged as soon as knee flexion permitted, usually by 2 to 3 weeks, and running on a treadmill commenced by 4 months. Return to pivoting and twisting sports was not recommended for 1 year.

Outcome Measures

Patients were evaluated preoperatively, intraoperatively, and postoperatively at 3, 6, 12, and 24 months. KT-1000 arthrometer testing at 134 N and manual maximum was performed intraoperatively, immediately before and immediately after reconstruction, as well as at all other time intervals. The KT-1000 arthrometer test was performed by the same trained and experienced examiner (S.C.) at all time intervals. At follow-up visits, a stockinette sleeve was placed over the knee to cover the surgical incisions in order to blind the examiner with respect to the graft type used. A questionnaire was completed by patients preoperatively and at all time intervals postoperatively. This instrument included questions regarding the ability to walk, climb stairs, kneel, squat, run, move laterally, jump, and cut. Patient response choices were no problem, some difficulty, extreme difficulty, or inability to perform the specified activity. Response choices to questions regarding frequency of pain and swelling were never, infrequently, frequently, or continuously. The level of overall pain, and pain with kneeling, were described as none, mild, moderate, or severe. Patients were asked to rate their knee from 0 to 100 preoperatively and at the 1-year and 2-year visits. Patients were also asked whether their knee was much better, slightly better, not better, or worse, at the 1- and 2-year follow-up visits. Physical examination at each time interval evaluated tenderness, effusion, pivot shift, and patellofemoral crepitus, which were recorded as none, mild, moderate, or severe. Sensory deficits were mapped out by the patient, and the surface area was calculated by multiplying the maximal length by the width. Range of motion was reported as the deficit in flexion and extension compared with the nonoperated leg. Extension deficit was measured as the prone heel-height difference in centimeters between the 2 extremities and then converted to degrees, assuming 1 cm equaling 1°, as shown by Sachs et al.[35]

Functional tests included a single-legged hop test, which compared the average of 3 maximal distance hops on the operated leg with the nonoperated leg. Strength testing was completed using a Biodex machine (Biodex Medical Systems Inc, Shirley, NY) at 6, 12, and 24 months. The parameters tested were flexion and extension, and abduction and adduction strength at 60 deg/s, 180 deg/s, and 300 deg/s. Internal and external rotation strength was tested at 60 deg/s, 120 deg/s, and 180 deg/s at 12 and 24 months. Outcome measures included the Lysholm,[25] Tegner,[40] and IKDC[17] rating scales, which were performed preoperatively and at 12 and 24 months. The Short Form-36 (SF-36) health status survey[42] was used at 12 and 24 months.

Statistical Analysis

A power analysis was performed with the assumption that a KT-1000 arthrometer manual maximum side-to-side difference of 1.5 mm was clinically significant. Using a

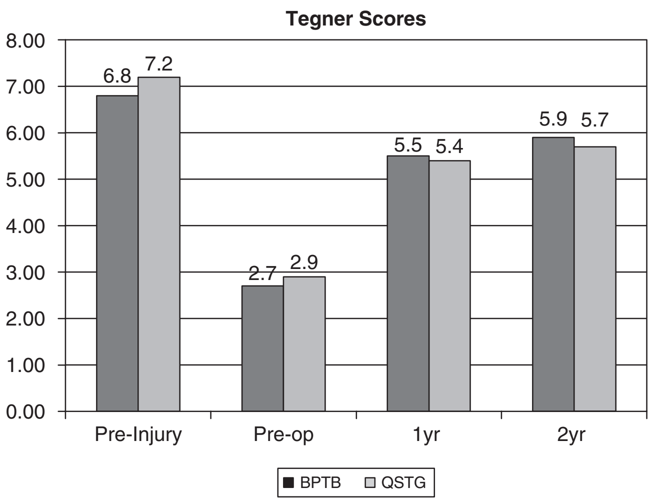

Figure 2. Tegner scores at each time point. Significant improvement was seen from preoperative scores to 1 year and 2 year scores in each group ($P < .0001$). There was a significant difference between the bone–patellar tendon–bone (BPTB) and quadrupled semitendinosus and gracilis tendon (QSTG) group preinjury scores ($P = .03$) but no difference at 1 year or 2 years. The mean Tegner scores did not return to their preinjury level for either group; however, 51% of patients in the BPTB and 26% in the QSTG group returned to the same or higher preoperative Tegner level ($P = .01$).

standard deviation of 2.0 mm, a power of 90%, and a confidence level of .05, a sample size of 39 patients per group was calculated. Assuming a patient loss of 15%, a minimum of 46 patients were required for each group.

Statistical analysis was performed using SAS software (SAS Institute, Cary, NC). The Wilcoxon rank sum test was used for evaluation of continuous variables (Tegner, Lysholm, KT-1000 arthrometer, range of motion, patient knee rating, and strength data). Categorical variables (pivot shift, anterior knee pain, crepitus, kneeling pain and frequency, activity ratings, pain level and frequency, and IKDC) were compared using the Pearson χ^2 test. The Wilcoxon signed rank test was used to compare changes within groups (KT-1000 arthrometer differences, pivot shift, Tegner, Lysholm, patient knee rating, and strength). The sign test was used to compare kneeling pain and difficulty with kneeling within each group between preoperative and 2-year postoperative time intervals.

RESULTS

Ninety-nine patients began the study (46 BPTB and 53 QSTG); the demographics are noted in Table 1. Two patients were lost to follow-up; 1 moved out of the country after his 3-month follow-up and could not be reached (QSTG), and 1 could not be reached for her 2-year examination (QSTG). At 1 year, she had a KT-1000 arthrometer manual maximum difference of 4 mm, an IKDC rating of nearly normal, and a subjective knee rating of 100. One patient (QSTG) ruptured his graft at 8 months and did not return for follow-up at 1 or 2 years. This left a total of 96 patients (97% follow-up) (46 BPTB and 50 QSTG) for the 2-year examination (mean, 26.2 months; range, 23 to 45). At the 3-month visit, there were 81 total patients (41 BPTB and 40 QSTG); at 6 months, 87 patients (39 BPTB and 48 QSTG); and at 1 year, 93 patients (43 BPTB and 50 QSTG). Three patients (2 BPTB and 1 QSTG) suffered an ACL tear of the contralateral knee during the study period; therefore, KT-1000 arthrometer data and single-legged hop test data were not included on those patients at the 2-year evaluation. One patient (BPTB) was diagnosed with rheumatoid arthritis between the 1-year and 2-year evaluations. Her overall IKDC grade dropped from nearly normal to severely abnormal because of pain with sedentary activity. We analyzed the data both with and without her 2-year data; because they did not change the outcome, her data were included.

Activity Ratings

Patient-reported activity ratings were similar at all time periods for ability to walk, climb, squat, run, and cut. At 2 years, a difference was noted in ability to kneel with 20% (9 patients) of the BPTB group having some difficulty kneeling as compared with 6% (3 patients) of the QSTG group ($P = .04$). No patient in either group stated that they had extreme difficulty or were unable to kneel. More patients in the QSTG group noted some difficulty with jumping activities at 2 years (17% [8 patients] as compared with 3% [1 patient] in the BPTB group [$P = .03$]). None of the respondents reported extreme difficulty or inability to jump; however, 8 patients in the BPTB group and 3 patients in the QSTG group stated that they had not attempted jumping activities. At 3 months, 17% (7 patients) of the BPTB group had frequent pain, whereas no patient in the QSTG group had frequent pain ($P = .01$); by 6 months and thereafter, this difference had resolved. Pain with kneeling was significantly higher in the BPTB group at 6 months, with 39% (12 patients) noting no pain, 45% (14 patients) with mild pain, 13% (4 patients) with moderate pain, and 3% (1 patient) with severe pain. In the QSTG group, 72% (28 patients) had no complaints of pain, 18% (7 patients) noted mild pain, 10% (4 patients) had moderate pain, and none complained of severe pain ($P = .03$). Eight patients in the BPTB group and 9 patients in the QSTG group failed to answer this question at the 6-month interval. At 1 and 2 years, the difference was no longer significant. No differences were noted with regard to swelling or giving way at any time interval.

No difference was noted between the BPTB and QSTG groups with either Lysholm or Tegner activity scores at 1 or 2 years (Figure 2). A significant improvement in Tegner and Lysholm activity was noted in both groups when preoperative data were compared with 1-year and 2-year data ($P < .0001$). There was a difference noted in ability to return to preinjury Tegner activity level at 2 years, with

TABLE 2
Outcome Measures[a]

	BPTB	QSTG	*P* Value
Lysholm (0-100)			
Preoperative	64	67	.42
1 y	95	96	.59
2 y	97	98	.58
KT-1000 arthrometer manual maximum difference (mm)			
Preoperative under anesthesia	8.3	8.3	.66
Postoperative under anesthesia	–0.8	–1.5	.06
3 mo	2.0	2.9	.04
6 mo	2.2	2.7	.33
1 y	2.5	2.7	.60
2 y	2.3	2.8	.20
Female KT-1000 arthrometer manual maximum difference (mm)			
3 mo	1.4	4.3	.01
6 mo	2.1	2.3	.81
1 y	1.8	3.3	.03
2 y	2.9	3.1	.51
Male KT-1000 arthrometer manual maximum difference (mm)			
3 mo	2.2	2.6	.46
6 mo	2.2	2.7	.35
1 y	2.9	2.6	.56
2 y	2.1	2.8	.14
Pivot shift (2 y)			.17
None (0)	42	39	
Glide (1+)	4	10	
Clunk (2+)	0	1	
Gross (3+)	0	0	
Sensory deficit (cm^2)			<.0001
None	8	43	
1-10	8	4	
10.5-20	9	0	
>20	21	3	

[a]BPTB, bone–patellar tendon–bone; QSTG, quadrupled semitendinosus and gracilis tendon.

51% (23 patients) of the BPTB group able to return to the same or higher level and only 26% (13 patients) of the QSTG group returning to the same or higher preinjury activity level (*P* = .01).

Clinical Assessment

KT-1000 arthrometer testing revealed significant differences only at 3 months. At 134 N, the mean difference was 1.4 mm in the BPTB group and 2.6 mm in the QSTG group (*P* = .01), with a manual maximum difference of 2.0 mm and 2.9 mm (*P* = .04), respectively. No differences in KT-1000 arthrometer testing were found at 6, 12, or 24 months (Table 2). In the BPTB group, a KT-1000 arthrometer manual maximum difference of 0 to 2 mm was found in 27 patients (61.4%), 3 to 5 mm in 15 patients (34.1%), and >5 mm in 2 patients (4.5%). With the QSTG group, a difference of 0 to 2 mm was seen in 23 patients (47%), 3 to 5 mm in 23 patients (47%), and >5 mm in 3 patients (6%). When KT-1000 arthrometer manual maximum differences were analyzed by sex, a difference was noted between the 2 groups only for women at 3 months and 12 months (Table 2). Pivot-shift grade did not differ between the groups (Table 2).

A sensory deficit was found in 83% of the BPTB group but in only 14% of the QSTG group (*P* < .0001) at 2 years (Table 2). Patellofemoral crepitus and effusion were not different between the groups at any time interval. Range of motion was good for both groups. At 2 years, the mean loss of flexion was 0.7° (range, 0° to 5°) in the BPTB group and 1.3° (range, 0° to 5°) in the QSTG group; the mean heel-height difference was 0.3 cm (range, 0 to 2 cm) and 0.1 cm (range, 0 to 3 cm), respectively. No statistical difference was noted for loss of motion at any time interval.

Single-legged hop test data revealed a significant difference at 1 year, with the BPTB group able to hop a mean distance of 82% of the nonoperated leg, whereas the QSTG group was able to jump 90% as far as the nonoperated leg (*P* = .04). By 2 years, this difference was no longer significant, with the BPTB group able to hop 94% of the uninvolved leg distance versus 97% for the QSTG group (*P* = .41).

Strength Testing

Knee extension strength improved at each time interval for both the BPTB and the QSTG groups (Table 3). The QSTG group had better extension strength at 6 months (180 deg/s, 300 deg/s), at 1 year (60 deg/s, 180 deg/s), and at 2 years (60 deg/s). Similarly, the BPTB group exhibited better flexion strength at 6 months (180 deg/s), at 1 year (180 deg/s, 300 deg/s), and at 2 years (180 deg/s) (Table 3). There were no differences between the 2 groups for abduction and adduction strength or for internal and external rotation strength at any time interval.

Patient Rating

A significant improvement in patient knee rating was seen from the preoperative to the 1-year rating, with the BPTB group improving from 50 to 88 and the QSTG group improving from 54 to 87 (*P* < .0001). At 2 years, the mean rating for each group was 93, which was a significant improvement from both the preoperative value and the 1-year value (*P* < .0001). There was no difference in knee rating between the groups at any time interval. When patients were asked whether, after surgery, their knee was much better, slightly better, not better, or worse, at 1 year, 86% (36 patients) in the BPTB group stated it was much better and 14% (6 patients) stated it was slightly better. In the QSTG group, 90% (44 patients) stated the knee was much better and 10% (5 patients) stated it was slightly better. One patient in each group failed to respond to the question. At 2 years, 93% (43 patients) of the BPTB group stated that the knee was much better and 7% (3 patients) said slightly better. At 2 years, 100% (50 patients) of the QSTG group felt their knee was much better. No patients rated their knee not better or worse at either the 1-year or 2-year interval. There were no differences between the groups at the 1-year or 2-year evaluations.

TABLE 3
Extension and Flexion Peak Torques Measured as a Percentage of the Opposite Leg[a]

	6 Months			1 Year			2 Years		
	BPTB	QSTG	*P* Value	BPTB	QSTG	*P* Value	BPTB	QSTG	*P* Value
Extension strength									
60 deg/s	N/A	N/A		70%	83%	.03	85%	92%	.04
180 deg/s	66%	78%	.002	80%	85%	.03	93%	96%	.18
300 deg/s	77%	86%	.01	85%	89%	.2	94%	96%	.18
Flexion strength									
60 deg/s	N/A	N/A		95%	89%	.12	99%	91%	.14
180 deg/s	96%	88%	.002	101%	88%	.0001	102%	90%	.0001
300 deg/s	96%	91%	.06	99%	91%	.01	96%	93%	.31

[a]BPTB, bone–patellar tendon–bone; QSTG, quadrupled semitendinosus and gracilis tendon.

IKDC

No differences were noted in the subjective assessment, symptoms, or overall IKDC for the 2 groups at 1 year or 2 years. There was a significant improvement from the preoperative grade to the 1-year and 2-year grades in each group ($P < .0001$). At 2 years, 91% of the BPTB group and 92% of the QSTG group were rated normal or nearly normal. One patient in the BPTB group had a rating of severely abnormal because of constant pain. This patient had developed rheumatoid arthritis between years 1 and 2 and had decreased from a nearly normal grade to a severely abnormal grade during that time.

SF-36 Short Form Health Survey

At 1 year, the BPTB group had higher scores for general health ($P = .036$) and vitality ($P = .021$) than the QSTG group. At 2 years, there were no differences noted between the 2 groups. If the men and women were analyzed separately, some differences were noted: men who underwent reconstruction with BPTB showed a trend toward higher physical function ($P = .055$); and women with a BPTB reconstruction had higher scores for social ($P = .038$), role emotional ($P = .027$), and mental functioning ($P = .036$). When BPTB and QSTG groups were combined, there was a significant increase in physical function between years 1 and 2 ($P < .001$).

Complications

One patient (QSTG) suffered a graft rupture at 8 months and elected not to have a revision. He did not return for his 1- or 2-year visits. One (QSTG) intra-articular coagulas-negative staphylococcal infection occurred that was treated by arthroscopic irrigation and intravenous antibiotics. A second patient (QSTG) received similar treatment for a presumed infection, but culture findings were negative. Both patients subsequently recovered well and were included in the final evaluation. There were 2 instances of intraoperative bioabsorbable screw fracture, 1 femoral (BPTB) and 1 tibial (BPTB); in each case, the bioabsorbable screw was replaced with a titanium screw without difficulty.

In summary, the BPTB group had better stability at 3 months; in women, this was also seen at 1 year. The SF-36 score was better at 1 year in the BPTB group for general health and vitality, and in women for social, role emotional, and mental function at 2 years. In the BPTB group, fewer patients had difficulty jumping, and a larger percentage of patients were able to return to the same or higher preinjury Tegner level. Flexion strength was better at all time intervals for the BPTB group. The QSTG group had fewer patients with pain at 3 months and fewer who had pain with kneeling at 6 months. At 1 year, the single-legged hop test results favored the QSTG group; at 2 years, fewer patients reported difficulty kneeling, and fewer patients had a sensory deficit at the incision site. Extension strength was better in the QSTG group at all time intervals.

DISCUSSION

The optimal graft for ACL reconstruction is still a subject of controversy. A number of prospective randomized studies using varying fixation constructs have been conducted to evaluate the differences between patellar tendon and hamstring grafts.[2,3,11-13,20,21,24,27,36] Most of these studies have failed to show significant differences in knee functional outcome scores. A meta-analysis by Freedman et al[14] showed better stability and a lower failure rate with patellar tendon grafts, but there was also a higher rate of manipulation under anesthesia and greater anterior knee pain as compared with the hamstring grafts. In another meta-analysis, Yunes et al[47] concluded that both grafts yielded good results, but that the patellar tendon graft provided a 20% greater chance to return to preinjury activity levels and also provided better stability as measured by the KT-1000 arthrometer. A more recent meta-analysis by Prodromos et al[33] contradicted the earlier studies and suggested that a quadrupled tendon hamstring graft provides better stability than a patellar tendon graft. Goldblatt et al,[15] in their recent meta-analysis, found that patients with BPTB grafts were more likely to have normal Lachman, normal pivot-shift, manual maximum KT-1000 arthrometer scores <3 mm, and less flexion loss. Patients with hamstring tendon autografts were less likely to experience kneeling pain, extension loss, and patellofemoral crepitus. Our study is an attempt to compare ACL reconstructions using BPTB and QSTG, while controlling for as many variables as possible, including type of fixation, so that any differences found should be attributable to the graft used.

Bioabsorbable interference screws have become popular for fixation of both BPTB and QSTG grafts. These screws have equivalent in vitro strength when compared with titanium screws[5,23,44] and have performed well in clinical trials.[28] Bioabsorbable screws have definite advantages over titanium screws when subsequent MRI scans are needed, and they may cause less damage to soft tissue grafts.[5] They are theoretically more advantageous in revision situations as well.[6] If the screw has resorbed and been replaced by bone, then redrilling of appropriate tunnels is simplified. The actual time for resorbtion may be longer than was originally thought,[34] and the theoretical advantage may not hold true because revision with a partially resorbed screw may actually be more difficult.

In this prospective randomized comparison, we found that both BPTB and QSTG autografts fixed with bioabsorbable interference screws performed well at 2-year follow-up. There was a significant improvement in both groups for the Tegner and Lysholm scales, as well as the patient-reported knee rating score. The IKDC subjective and overall scores were also noted to significantly improve from preoperative to 2-year evaluation. We did not find a difference in our primary outcome objective, the KT-1000 arthrometer manual maximum difference, between those patients reconstructed with BPTB or QSTG autografts at 2 years. There was a statistically significant difference at 3 months, with the BPTB group achieving better stability than the QSTG group. Although the BPTB group was tighter at each interval, the difference was not statistically significant other than at 3 months. It is difficult to compare KT-1000 arthrometer data with other studies in the literature. Different examiners are used, and many studies do not report manual maximum differences. Furthermore, reporting of only mean differences may be misleading because a negative KT-1000 arthrometer value may negate a loose KT-1000 arthrometer value; the overall mean value may be acceptable, yet in reality both patients are failures. In our experience, patients with negative KT-1000 arthrometer values have usually lost motion, which leads to a poor result. In this study, no patients had negative manual maximum differences, and there were only minimal range of motion losses in both groups. Aune et al[3] reported mean KT-1000 arthrometer manual maximum differences of 2.7 mm for both the patellar tendon and hamstring groups, but they did not report the distribution of the values. Aglietti et al[2] used 134-N KT-1000 arthrometer results and reported mean differences of 1.95 mm and 2.2 mm, and ≤2 mm for 65% and 57% of patients with patellar tendon and hamstring grafts, respectively. Our manual maximum results of 2.3 mm for the patellar tendon group and 2.9 mm for the hamstring group, with 61% and 47%, respectively, ≤2 mm are comparable. Other prospective randomized trials[20,27,36] have reported KT-1000 arthrometer manual maximum data with tighter values; in some cases, however, knees with negative KT-1000 arthrometer values may have influenced the overall result. Longer follow-up will be required to determine if the KT-1000 arthrometer values are stable over time and how they relate to overall outcomes.

Although we had relatively few women enrolled in this study, we did find that the women who underwent reconstruction with QSTG grafts had greater laxity at 3 months and 1 year than did those with BPTB grafts. Increased laxity in women with a hamstring graft fixed with interference screws has been described by others.[10,29] In the study by Corry et al,[10] a single metal interference screw was used on the tibial side. They surmised that the decreased bone quality of the tibia may allow for graft slippage, which may explain the increased laxity seen in women. To improve the fixation on the tibial side, we elected to use bicortical fixation in the tibia with 2 screws. This has been shown to increase the maximum load to failure, increase stiffness, decrease graft slippage, and have good clinical outcomes.[7,8] Despite the apparent improvement in tibial fixation, we still found increased laxity in women at the 3-month and 1-year time intervals.

For patient-reported activity levels such as ability to climb stairs, kneel, squat, run, and jump, we found only differences in the ability to kneel and jump. More patients in the QSTG group noted some difficulty with jumping, but none of the respondents stated that they had extreme difficulty or were unable to jump. This difference in the subjective response to the question regarding ability to jump is different from what was measured objectively by the single-legged hop test, for which no significant difference was found between the groups. We found that the respondents who noted some difficulty with jumping had a significantly lower hop test result than those stating no difficulty, but the number of patients was small and did not decrease the otherwise good hop-test results. Why more patients who underwent QSTG grafts reported difficulty jumping is unclear. Many authors have noted increased kneeling pain[2,3,11-13,24] with the patellar tendon group after ACL reconstruction, and kneeling pain has often been considered one of the main drawbacks to using a BPTB graft. We found that more patients in the BPTB group noted some difficulty with kneeling, although none reported that they had extreme difficulty or were unable to kneel. Interestingly, at 2 years, when patients were asked about pain with kneeling rather than difficulty with kneeling, there was no difference between the groups. We actually found a decrease in kneeling pain and an increased ability to kneel from the preoperative evaluation to the 2-year evaluation in both groups. Our patellar tendon harvest technique uses an incision that is to the medial side of the patellar tendon and tibial tubercle, which may have minimized the pain associated with kneeling and accounted for the fact that there was no statistical difference in patient reports of kneeling pain between the 2 groups. There was a significant difference in sensory deficit about the knee in the BPTB group, which has also been noted in other studies.[2]

The Tegner and Lysholm scores were not different between the 2 groups at 2-year follow-up. The mean Tegner score did not return to the preoperative level for either group. We did find that a greater percentage of the BPTB patients were able to return to the same or higher preoperative Tegner level than the QSTG group, but the QSTG group started out at a higher preoperative level, which may account for this difference. Other investigators have also noted that many patients are not able to return to their preinjury level of sports participation, irrespective of graft type.[2,11,13,24,27,36] This may in part be related to the surgery, fear of reinjury, or the natural decrease in activity level over time.

Many authors have reported decreased range of motion in the patellar tendon group.[12,13,20,21,36] Loss of motion, especially extension, has been linked to increased anterior knee pain.[22,35] We did not see clinically significant loss of motion in either group or differences between the BPTB and QSTG groups. We believe strongly that maintaining full range of motion is important for maximal functional recovery. Range of motion exercises were emphasized preoperatively and at all postoperative visits. Flexion and extension exercises were begun the same day as surgery, which may have accounted for the minimal decreases in range of motion seen.

Of the randomized controlled trials (RCTs) comparing quadriceps and hamstring strength, some have found no difference between the graft types.[2,21] Others have found a difference, with hamstring muscle weakness noted in the hamstring tendon group.[3,13,27] We found a significant decrease in hamstring muscle strength in the QSTG group, as well as a decrease in the quadriceps muscle strength in the BPTB group. Just how the strength deficit affects return to sports and risk of reinjury is unclear. There is some concern with respect to further weakening of the hamstring muscle group with use of a QSTG graft in the female athlete, who may already have a low quadriceps-to-hamstring peak torque ratio.[18]

Four RCTs with minimum 2-year follow-up have used the same fixation for each graft.[11,24,27,36] In each of these studies, metal interference screws were used. No differences were noted in laxity,[11,24,27,36] Lysholm,[11,24] Tegner,[11,24] single-legged hop test,[11,24] or IKDC scores.[11,24,27] Shaieb et al[36] found increased anterior knee pain in the BPTB group but also noted a significant difference in range of motion between the 2 groups, with the BPTB group having decreased motion. Laxdal et al[24] found a significant difference in knee walking ability but no difference in knee sensation loss. They did not show a significant difference in range of motion between groups but found 34% of the BPTB group and 16% of the QSTG group had a loss of ≥5° or more of extension, and 63% and 68%, respectively, lost ≥5° of flexion. Ejerhed et al[11] found no difference in strength measurements, knee sensory loss, or anterior knee pain. They also found no difference in range of motion, but they did note a loss of extension of ≥5° in 41% of the BPTB group and 26% of the hamstring group, and flexion loss of ≥5° in 41% of the BPTB group and 71% of the hamstring group. We found no differences in kneeling pain between our 2 groups, and although there was a difference in patient-reported ability to kneel, no patient in either group complained of extreme difficulty or was unable to kneel. We did not ask patients to perform a knee-walking maneuver, as has been done by others,[11,24] which may have further differentiated the 2 groups. We did find that loss of sensation was significantly greater in the BPTB group. We found only small range of motion deficits in our study, with 13% of the BPTB group and 26% of the QSTG group losing ≥5° of flexion; no patients in either group lost ≥5° of extension.

In one comparative study in the literature, bioabsorbable screws were used for both BPTB and QSTG grafts.[41] That study used a matched cohort rather than a prospectively randomized design, and the tibial fixation for the hamstring grafts was supplemented by sutures tied over a bone bridge in addition to the interference screw. Contrary to all of the RCTs in the literature, that study found that the hamstring grafts were superior in nearly all outcome measures (KT-1000 arthrometer, IKDC, single-legged hop test, Lysholm score, thigh atrophy, and patellofemoral crepitus). Although we also used bioabsorbable screws for fixation for both grafts, our results, like the other RCTs, did not demonstrate superiority of the hamstring grafts in most of these same outcome measures.

The strengths of this study are that it is prospective and randomized, thereby minimizing susceptibility (selection) bias. A single surgeon with experience using both techniques performed all surgeries. The surgical technique was the same, and all grafts were fixed with bioabsorbable interference screws. Rehabilitation was consistent for all patients and directed by the same therapist. Follow-up was conducted by a single examiner, and a stockinette sleeve was used to cover the knee in an attempt to keep the examiner blinded. There was a high rate of follow-up, thereby minimizing transfer bias.

One of the weaknesses of the study is that there was a small difference noted in preoperative Tegner activity scores, which means that despite randomization, the groups may not have been completely equivalent. We did not find a difference when preoperative activity was evaluated by the IKDC activity level or Lysholm score, thus we are confident that the small difference in Tegner scores did not significantly bias the results. The power analysis was performed to look for KT-1000 arthrometer differences between the groups. It is possible that clinically significant differences in other outcomes were not detected because of the inadequate sample size. During the consent process, we explained to the patients that the literature at that time[1,10,26,30] suggested that there were some slight differences in outcome between the 2 grafts. The QSTG group reportedly had less kneeling pain, and the BPTB group had better stability, but no differences were noted in overall functional outcome between the groups. It is likely that patients at high risk for kneeling pain or those concerned with cosmesis chose QSTG grafts, while those who put a premium on stability, possibly the more athletic patients, chose BPTB grafts. This appears to be borne out by the preinjury Tegner scores of the nonstudy group, in which the BPTB group had higher mean preinjury Tegner scores and the QSTG group had lower scores than those seen in the study group.

We found that 53% of the patients eligible for the study opted not to be randomized. This is not uncommon for a surgical trial in the United States. This is one of only 2 prospective randomized trials with 2-year follow-up comparing BPTB and QSTG grafts performed in the United States.[36] The other studies in the literature do not provide the number of patients who refused randomization, so we are unable to evaluate the effect that this may have had on the patient selection.

CONCLUSIONS

Reconstructing the ACL using either the BPTB or QSTG graft can lead to success. We found an overall patient knee rating of 93 in both groups and a normal or nearly normal IKDC grade in 91% of the BPTB and 92% of the QSTG-reconstructed patients. The BPTB graft led to an increase in anterior knee sensory deficit and complaints of difficulty kneeling. Strength differences were noted, with the BPTB group losing some quadriceps strength and the QSTG group losing some hamstring strength. There were fewer reports of difficulty with jumping in the BPTB group. We found similar outcomes at 2 years for KT-1000 arthrometer manual maximum difference, Lysholm score, Tegner level, and IKDC grade, all of which were improved compared with the preoperative evaluation.

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